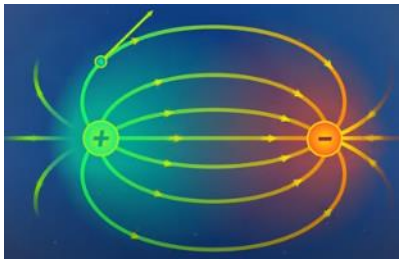
EE4047 – High Voltage Techniques I

Continuation to:
Electrostatic Fields and Planar Electrode Systems



Week 6.1 – Monday Lesson

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OUTLINE

2 Continuation to: Electrostatic Field in Planar Electrode Systems

Example-1:

In a planar electrode system, electrode gap is a=2 mm, electrode surface area $S = 4\,cm^2$, $\varepsilon_r = 1\,(air)$ and the applied voltage is U=5 kV. Calculate the capacity, electric charge, electric field strength, electric flux density and the maximum voltage that can be applied if the dielectric strenght of the insulation material is 30 kV/cm. ($\varepsilon_0 = 8{,}854.10^{-12}\,\frac{F}{m}$)

Formulas:

$$D = \varepsilon E$$

$$E = \frac{U}{a}$$

$$Q = C.U$$

$$C = \frac{\varepsilon S}{a}$$

Solution: Capacity of the electrode system:

$$C = \frac{\varepsilon S}{a} = \frac{\varepsilon_0 \varepsilon_r S}{a} = \frac{8{,}854.10^{-12}.1.4.10^{-4}}{2.10^{-3}} = \frac{F.m^2}{m.m} = 1{,}7708.10^{-12}\,F = 1.7708\,pF$$

Electric charge:

$$Q = C.U = 1{,}7708.10^{-12}.5.10^{3} = 8.854.10^{-9}\,C = 8.854\,nC$$

Electric field strength:

$$E = \frac{U}{a} = \frac{5\,kV}{2.10^{-1}} = 25\,kV/cm = 2500\,\frac{kV}{m} = 25.10^{5}\,\frac{V}{m}$$

Example-1:

In a planar electrode system, electrode gap is a=2 mm, electrode surface area $S = 4\,cm^2$, $\varepsilon_r = 1\,(air)$ and the applied voltage is U=5 kV. Calculate the capacity, electric charge, electric field strength, electric flux density and the maximum voltage that can be applied if the dielectric strenght of the insulation material is 30 kV/cm. ($\varepsilon_0 = 8{,}854.10^{-12}\frac{F}{m}$)

Continuation to Solution: Electric flux density:

$$D = \varepsilon E = \varepsilon_0 \varepsilon_R E = 8{,}854.10^{-12}.1.25.10^5 = 2.2135.10^{-5}\,C/m^2$$

The breakdown voltage:

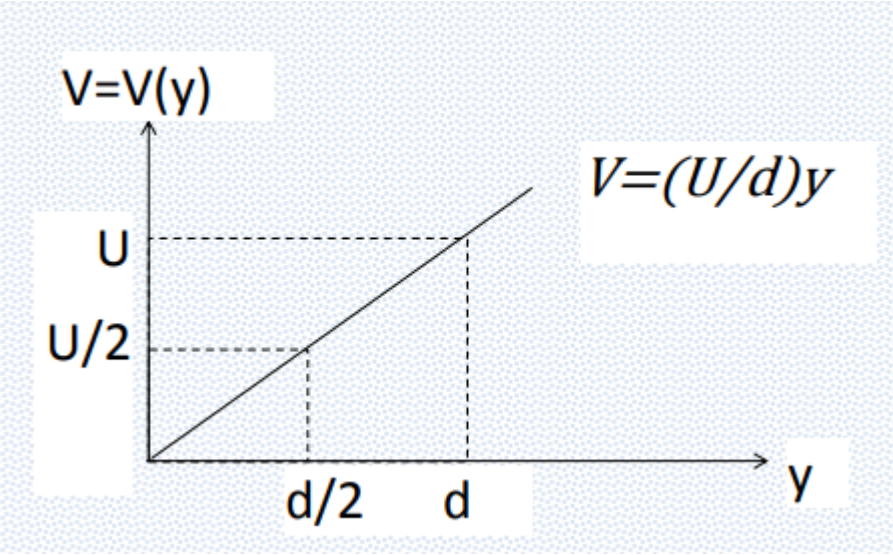
$$U_d = E_d.a = 30\frac{kV}{cm}.2.10^{-1}cm = 6\,kV$$

The maximum voltage that can be applied:

$$U < U_d = 6\,kV$$

ELECTROSTATIC FIELD IN PLANAR ELECTRODE SYSTEMS

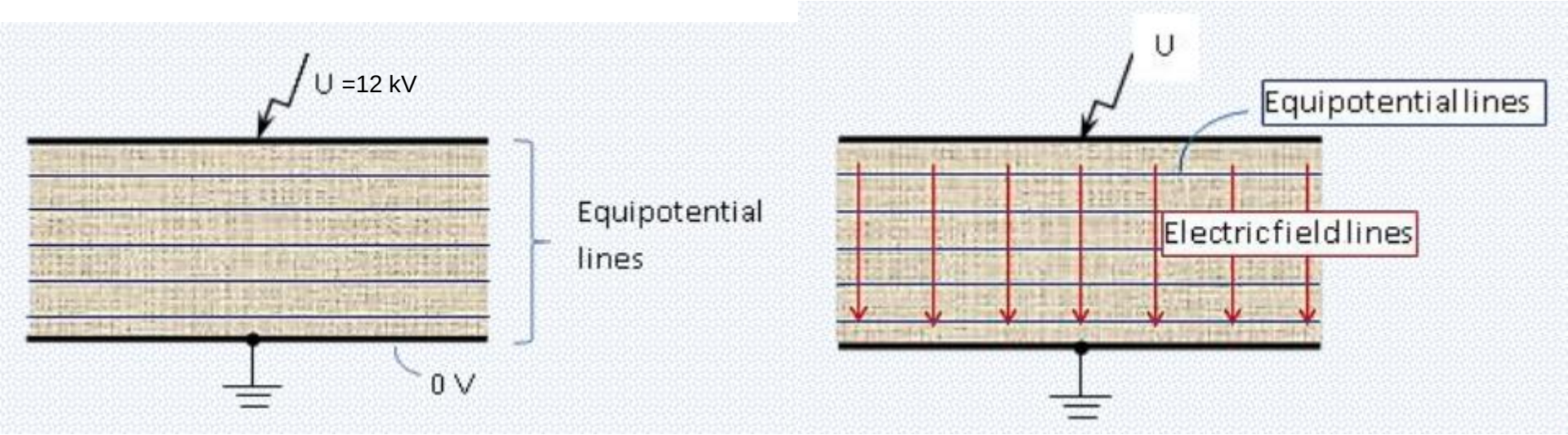
Change of voltage with distance (d):



2 ELECTROSTATIC FIELD IN PLANAR ELECTRODE SYSTEMS

Equipotential lines, the lines on which every point has the same potential.

Electric field lines are **ALWAYS perpendicular to** equipotential lines.



END OF THE LESSON

Any questions?

Check out our playlists on Youtube:

High Voltage Technique: https://www.youtube.com/playlist?list=PLLZI7WaEmZi04KMWVURGlU34W3LoNZyWX

Insulators: https://www.youtube.com/playlist?list=PLLZI7WaEmZi1ReLGq0qfP1VjRqxDJi5kA

Lightning Impulse Voltage: https://www.youtube.com/playlist?list=PLLZI7WaEmZi2isWOB-zBbb6LNplH28byQ